



HOCHSCHULE HEILBRONN

Metagenomics for Pathogen Detection

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experimenta

Das Science Center

Disclosure

I disclose the following relevant financial or non-financial relationships:

- Presenter is author of the metagenomics software “Genestrip”
(Genestrip is free for non-commercial and not commercially used.)

Mentioned Names of Organizations and Products

- “HHN” (Hochschule Heilbronn / Heilbronn University) – Speaker’s institution (state’s authority)
- “Experimenta” – a “science center” in Heilbronn Germany, in cooperation with HHN (non-commercial)
- “Fraunhofer” – German research institution (partially funded by state)
- “Noscendo” – German lab using metagenomics for diagnostic findings (commercial)
- “Nanopore” – manufacturer of sequencing devices (commercial)
- “krakenUniq” – free software for metagenomics (USA)
- “Genestrip” – partially free software for metagenomics, speaker is author of this software

Learning Objectives and Key Take-Aways

Understanding metagenomics with regard to:

1. Its basics and what process steps a metagenomic analysis consists of
2. How it compares to existing diagnostic tests for pathogens
3. Application, practicality and entry cost
4. Current state of research, author's research and example results
5. The future of metagenomic applications

1. Basics

Existing Diagnostic Tests for Pathogens

- **PCR**
 - **Antibody**
 - **Leukocyte Transformation (LTT)**
 - **Microscopy**
 - **Biopsy in conjunction with PCR or Microscopy**
- ⇒ Only PCR and Microscopy show presence directly!

Testing Problems:

- PCR is on genus- or species-level ⇒ requires an upfront guess of the infection!
- Immune-related tests may fail under immune suppression ⇒ also require an upfront guess of the infection
- Microscopy may be misleading ⇒ “What do we actually see there?”

Metagenomics and Whole Genome Sequencing

- Idea: Sequence and analyze **ALL of the DNA from a sample**
 - as all organisms in sample leave respective traces behind
- This has become relatively “easy” as the latest sequencing devices are
 - small,
 - inexpensive (less than 3000 USD),
 - enabled for ubiquitous use and
 - offer sufficient quality for metagenomics.

The Raw Sequencing Data as Produced by Sequencer

Extract of a „fastq“ file – a so-called “**read**”:

```
@SRR001666.1 071112_SLXA-EAS1_s_7:5:1:817:345 length=36
GGGTGATGGCCGCTGCCGATGGCGTCAAATCCCACC
+SRR001666.1 071112_SLXA-EAS1_s_7:5:1:817:345 length=36
IIIIIIIIIIIIIIIIIIIIIIIIIIIIIIII9IG9IC
```

Fastq files can become huge (by running the sequencer for long enough time on a sample)

- > 50 GB with billions of reads
- Depending on device, read length varies between a few 100 base pairs (“short reads”) to a few 1000 base pairs (“long reads”)
- But you never get the whole genome in one read – only a tiny fraction (except for viruses maybe)
- E.g., a bacterial genome comprises of millions of base pairs

Where is the DNA from?

- From potentially any organism in the sample!
- E.g., a saliva sample mainly contains human DNA
- But 1% to 5% are non-human
 - From food (plants, meat), gut and mouth bacteria and viruses etc.
- And similarly: Blood is typically not sterile!

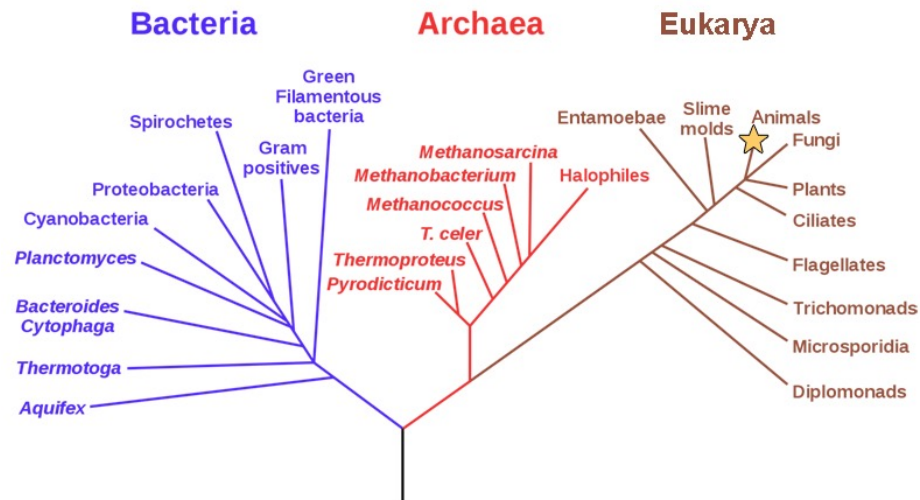
Matching Reads with Genomes

- A collection of genomes of over 155.000 organisms is at hand!
- All of it is **ready for download from the „National Center for Biotechnology Information”** (NCBI, a US authority):
 - <https://www.ncbi.nlm.nih.gov/home/download/>
 - Including an up-to-date **taxonomy tree** covering all known organisms
- Approach: **Find the matching genome for any read from a sample!**
 - Unfortunately, it's a search with an enormous computational overhead
 - It mandates specialized software and (often) large computers

Taxonomy Tree from Biology – the „Tree of Life“

Phylogenetic Tree of Life

★ = You are here



- In bioinformatics, every organism has its own numeric identifier – the „**tax id**“
 - E.g. „homo sapiens“ has 9606
- Every node of the entire taxonomy tree is organized via tax ids

Source: <https://courses.lumenlearning.com/wm-nmbiology1/>

Downloading Genomes from the NCBI...

Example link:

<https://www.ncbi.nlm.nih.gov/data-hub/taxonomy/tree/?taxon=64895>

- DNA file in the so-called **fasta format**:

```
>CP001568.1 Borrelia burgdorferi Bol26 plasmid Bol26_cp26, complete sequence
TTAAAAACTTTTCTATTGGATAGATTTTATACAAAGAAGGTAATAATGTATAACAACAATATTTTATTTCTGGCAAGGT
ACAAGGTGTTGGTTTTAGATTTTTTACAGAGCAAATAGCAAATAATATGAACTAAAAGGATTTGTAAAAAATCTCAACG
ATGGAAGGGTAGAAATTGTAGCTTTCTTTAATACTAAAGAACAAATGAAAAAATTTGAAAAATTATTAAATGGGAATAAG
TATTCAAACATTAAAAACATTGAAAAAATAGTTTTAGATGAAAATTATCCTTTTCAATTTAATGATTTTAAAATTTATTA
TTAGGGCTTGCCTCTCGTTTAACAAGTACCTTAACCTTATTTTTTTGGTTTAATATTGTGCACATAAGAATTGTTATTCTT
ATAGCAAGACACACTAGTTCTAAAAAAATGTTTCGACTTTAAATTTCAAAAACCTCTAAAGACTTTCTGTTTCTACAAAAA
...
```

„Classification“ of Reads

A bioinformatics term which means:

Assigning the “correct” tax id to every read from a fastq file

E.g., which organism is the following sequence most likely from?

GGGTGATGGCCGCTGCCGATGGCGTCAAATCCCACC

Software for Fast and Reliable Read Classification

General Computational Steps:

1. Breakup genomes into small parts of fixed size k – so-called **k-mers**
 - In most cases one uses $k = 31$ (for technical reasons), e.g., this is a 31-mer:
GGGTGATGGCCGCTGCCGATGGCGTCAAATC
2. Create a registry for all k-mers of candidate genomes
 - It's a huge table with entries of the sort
<k-mer, tax id>
 - In bioinformatics this registry is usually called a „**database**“
3. Problem: Due to bioevolution, many k-mers are identical between closely related organisms A and B
 - So, a database must account for this by storing the tax id of the **least common ancestor of A and B** for such a k-mer

2. Metagenomics versus other Tests

Diagnostic Steps using Metagenomics



1. Physician / Patient: Extracts sample and sends it to lab
2. Lab: Performs DNA sequencing on sample to get reads in fastq file
3. Computational analysis: Classify reads from fastq file
4. Scientist / Physician: Interpret resulting statistics
 - Which organisms were found?
 - How many reads classified per organism?
 - Which may imply: Which infections are present?

Metagenomics versus PCR (1)

PCR:

- PCR requires the definition of a **primer** – a very short but highly specific sub-sequence from an organism's genome
 - Usually determined for the taxonomic rank “genus” or “species”
- PCR “searches” for the primer in the sample
- Highly specific and sensitive, but:
 - **Other parts of the organism's genome are ignored and missed!**
 - **Other organisms (not containing the primer in their genome) are (usually) missed entirely!**

Metagenomics versus PCR (2)

Metagenomics:

- **All of the DNA** (in the sample) will be analyzed
- **All potential pathogens** in the sample can be detected with a single test
- No need to define or care about primers
- No preference to any organism, (typically) no amplification done
- **Specificity: Usually down to species or even strain level!**
- **Sensitivity:** Current research indicates: **As good as the most sensitive PCR techniques, if not better!**

Metagenomics versus 16S Sequencing

- 16S is an established sequencing method based on “older” sequencing devices
- **It focuses on a special genomic region called “16S” present in all bacteria (only)**
- The region is amplified (many copies are produced) using PCR and then sequenced

Cons:

- Amplification complicates sample preparation
- Works only for bacteria, but nothing else
- Can be too unspecific as related bacteria may have (almost) identical 16S regions
⇒ Often, bacteria species cannot be discriminated

Pros:

- A well-established analysis pipeline!
- A lot less data, and so a lot less compute required

3. Application of Metagenomics

Software Tools for Read Classification (1)

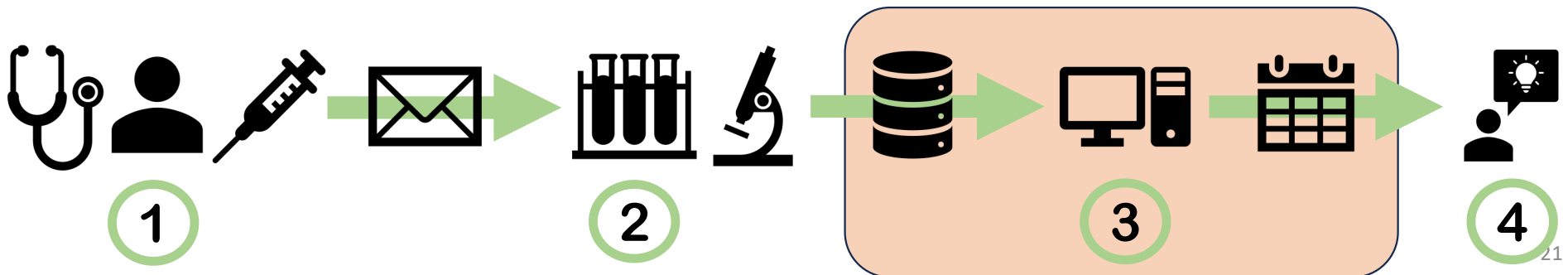
- Many tools from bioinformatics exist
- All of them are hard to use for laymen
- Some more efficient ones (like “kraken2”) are too imprecise for diagnostics and therefore unsuitable
- „[krakenUniq](#)“ is currently the “tool of choice” for potential diagnostics

Pros of krakenUniq:

- (More) Precise and fast read classification
- Decent result reports
- Open source and free

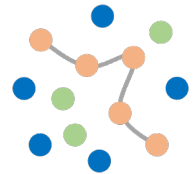
Cons of krakenUniq :

- Unix- and command-line based
- Enormous memory requirements (typically > 200 GB of RAM)
- krakenUniq-Databases are hard to “manage”
- Standard databases are often unsuitable for “organisms of interest”



Software Tools for Read Classification (2)

Due to the drawbacks of krakenUniq, I built my own software "[Genestrip](#)" for this very purpose



Pros of Genestrip:

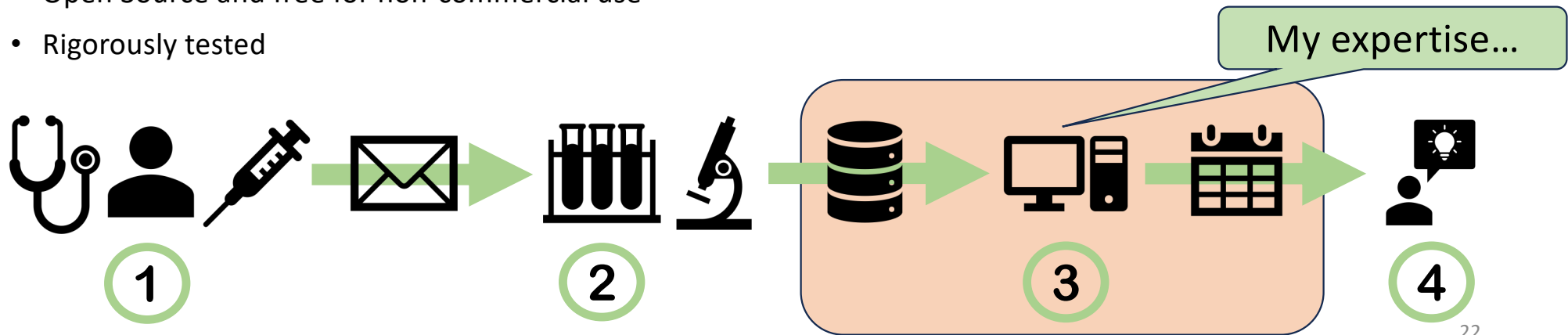
- As precise as krakenUniq
- Easier to use with an optional Web interface
- Runs on any regular PC (and Windows) and online
- A lot less memory required
- Databases are easier to build and can be small
- Open Source and free for non-commercial use
- Rigorously tested

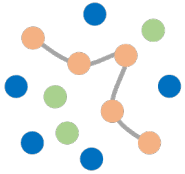
Cons of Genestrip:

- Tiny user community
- Slightly slower than krakenUniq

Neutral:

- Databases typically comprise a pre-selected set of organisms of interest





A Glance at some Genestrip-Databases

Potentially tick-borne bacteria:

<https://genestrip.it.hs-heilbronn.de/gweb/dbinfo?lan=en&dbid=4>

Human viruses:

<https://genestrip.it.hs-heilbronn.de/gweb/dbinfo?lan=en&dbid=5>

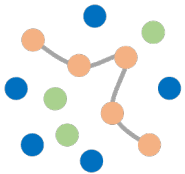
Babesia:

<https://genestrip.it.hs-heilbronn.de/gweb/dbinfo?lan=en&dbid=1>

Human Parasites:

<https://genestrip.it.hs-heilbronn.de/gweb/dbinfo?lan=en&dbid=6>

(To my knowledge, the only parasite database of this kind.)



Example Application: Pathogens in Field-Collected Ticks

- In this Nature publication, 8 ticks from different US-regions were analyzed via “Nanopore” sequencing:

<https://www.nature.com/articles/s41598-023-37134-9>

- All related fastq files are available for download
- I analyzed them too with Genestrip:
 - All of the publication results were recovered, down to species-level and below!
 - Let’s take a look at “Tick 3”:
<https://genestrip.it.hs-heilbronn.de/gweb/view?lan=en&jobid=94>
 - It is safe to say that:

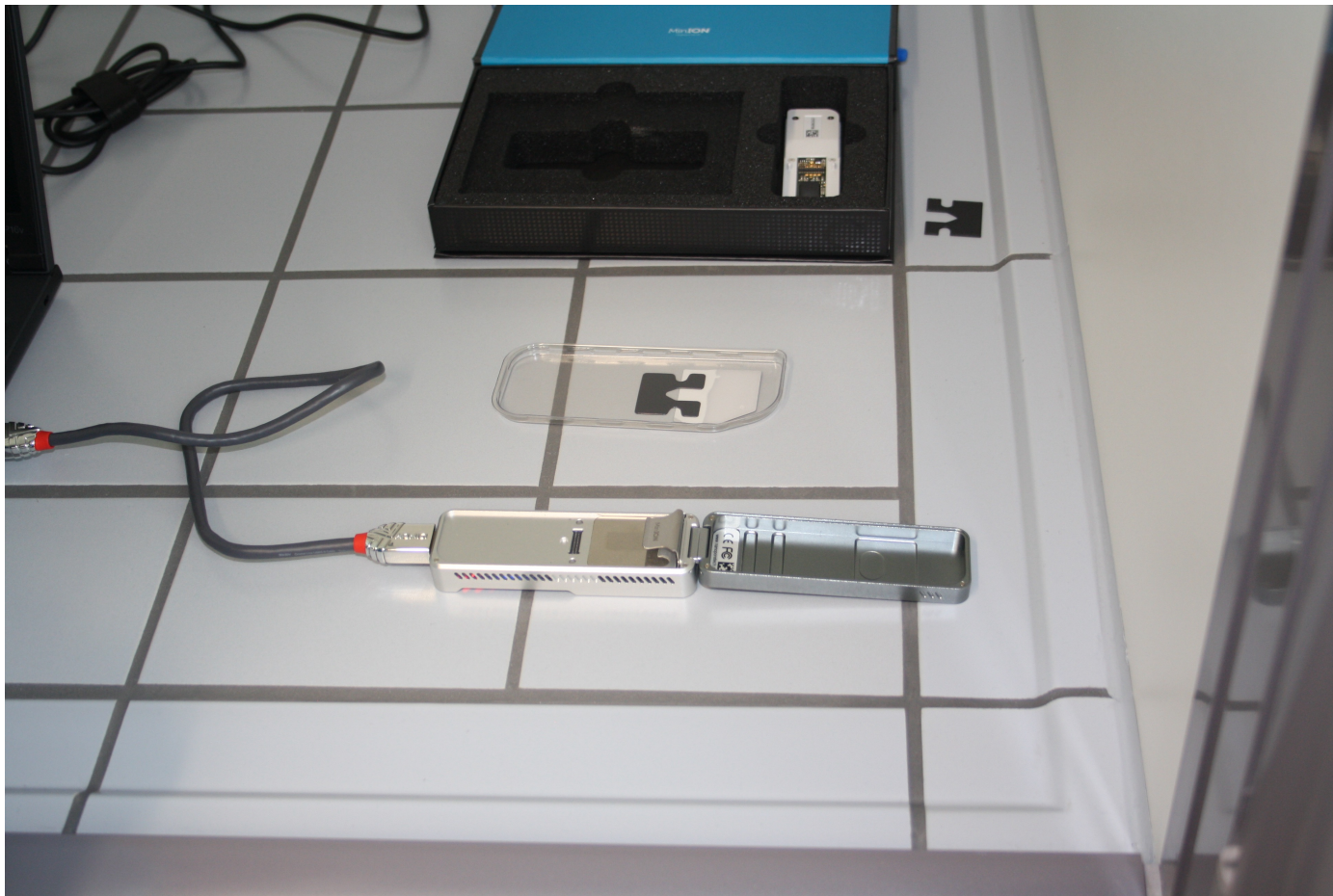
Metagenomics would be a great approach for tick surveillance!

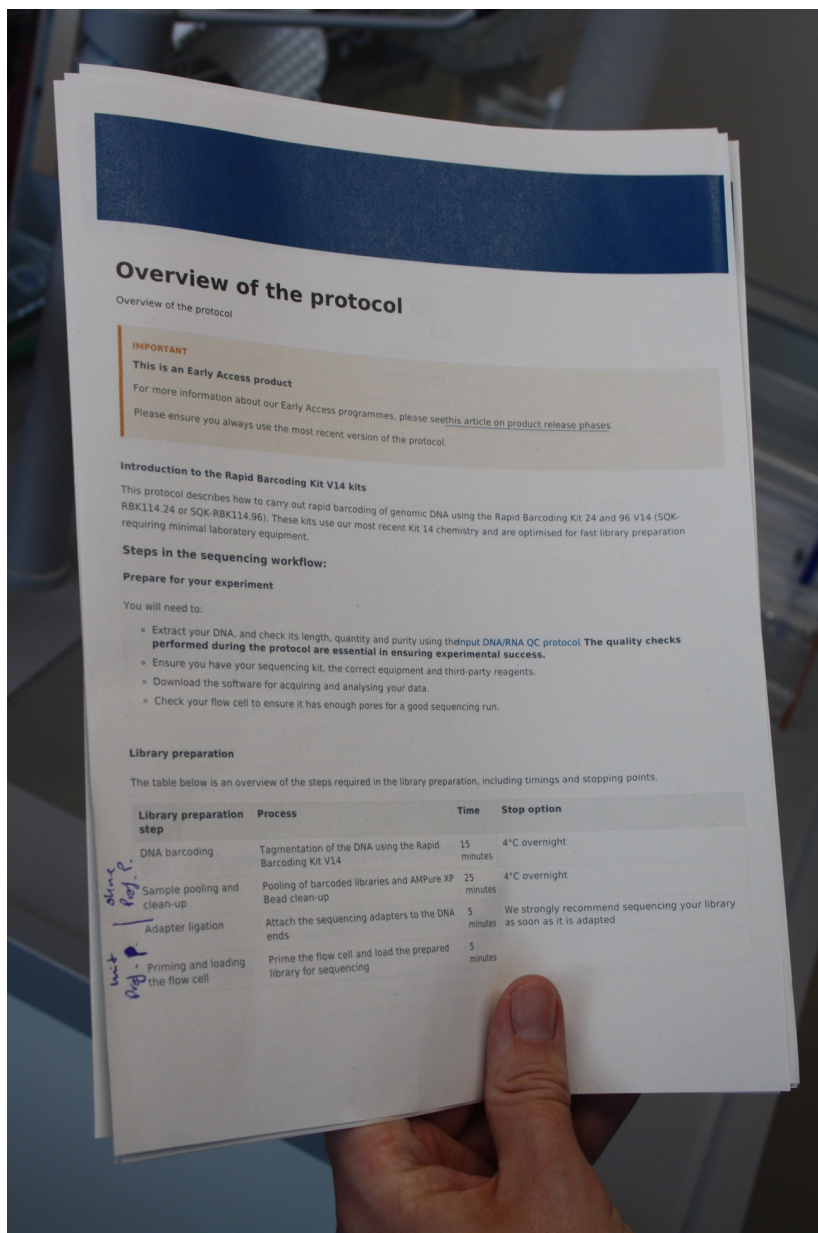
“Oxford Nanopore” is a producer of DNA-sequencing devices (with some very small models at very “low cost”). The terms “Nanopore” and “ONT” are widely used in scientific publications to indicate the use of such a device.

August 2024: “Live” Sequencing of Soil Samples

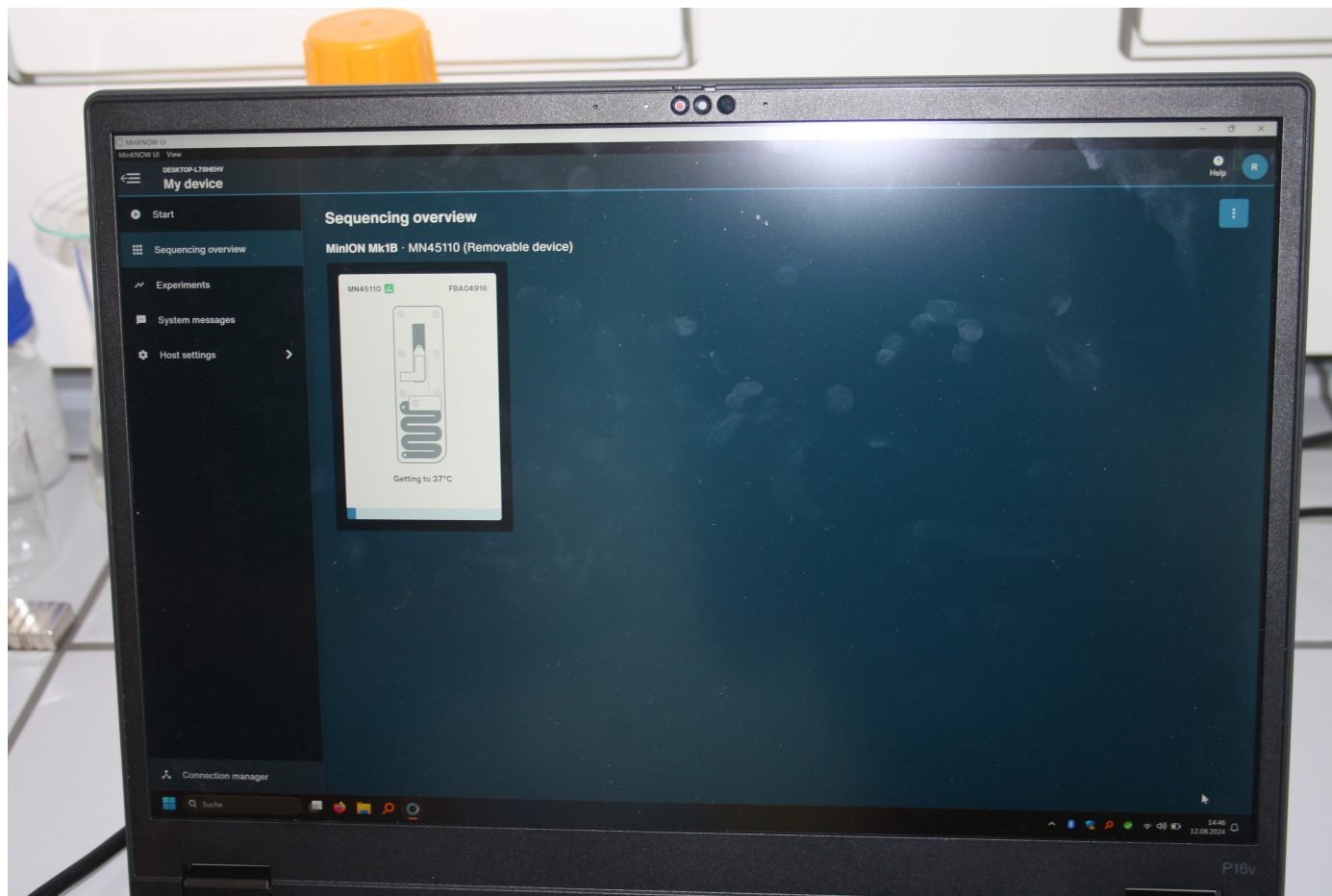
- In cooperation with Dr. Friedrich in an S1-Biolab of the „Experimenta“
- Found a fungus harmful to grape vine in some soil samples
- Base on a dedicated [Genestrip database compiled for grape vine fungi](#)
- Findings (partially) confirmed via PCR











The Entry Cost for Metagenomics

Basis: Medical Lab with standard equipment (protection class 3)

- Smallest sufficient sequencing device: 2000 Euros
- PC for Analysis: 3000 Euros
- DNA preparation kit for blood samples (for 50 samples): 200 Euros
- Consumable material for a sequencing device (a so-called “flowcell”): 60 Euros
 - Depending on the manufacturer, a flowcell can be reused up to 10 times

But many staff hours (incurring cost):

- Manual sample preparation takes about 4 to 6 hours for trained staff,
- But using a technique called “barcoding” up to 24 samples can be analyzed in one go
- Analysis software must be used adequately

4. Research and Existing Solutions

Analysis of Ticks (2023)

Seen before: **Nature** Publication regarding ticks

<https://www.nature.com/articles/s41598-023-37134-9>

8 field-collected ticks from different US regions analyzed via Nanopore sequencing

- Borrelia detected down to species level
- Same for Rickettsia and Anaplasma!

(Example results shown before.)

Pathogens in Stray Dogs from Bhutan (2024)

<https://www.sciencedirect.com/science/article/pii/S2352771424001654>

- Analyzed dog blood using Nanopore sequencing
- Pathogens found:
 - Mycoplasma
 - Bartonella
 - Babesia
 - Filaria

(Apparently, they did not look for Borrelia.)

Bartonella in Deer (in China, 2024)

<https://pmc.ncbi.nlm.nih.gov/articles/PMC11218516/>

- Analyzed deer blood using Nanopore sequencing
- Confirmation using PCR
- „Nanopore Sequencing“ has “higher sensitivity than PCR”
 - “... In this study, we adopted Nanopore sequence technology on the qPCR-positive samples. **The results of Nanopore sequencing can prove the result of qPCR. The sensitivity of Nanopore sequencing is higher** evidenced by the *Bartonella* reads that were detected in samples with Ct values higher than 35 (up to 37–38). ... “

Babesia in Dogs (2024)

<https://pubmed.ncbi.nlm.nih.gov/37837372/>

- Analyzed dog blood using Nanopore sequencing (100 dogs)

„... For two common haemoparasites, **nanopore sequencing performed as well as qPCR** (...), whilst also detecting one pathogen, Hepatozoon felis, missed by the other techniques. ...“

„... The long-reads obtained by nanopore sequencing provide an improved species-level taxonomic resolution whilst the method's broad applicability mean it can be used to explore apicomplexan communities ...“

Sequencing for Sepsis at “Fraunhofer”

<https://www.igb.fraunhofer.de/de/forschung/in-vitro-diagnostics/next-generation-diagnostics/infektionsdiagnostik-mit-next-generation-sequencing.html>

- **Already in use in hospitals**, especially in large facilities in the US
 - Faster detection of bacterium including potential resistance genes
 - To give best matching antibiotics early on
 - Very successful...

Noscendo – German (Startup-)Lab for Metagenomic Diagnostics

<https://noscendo.com/>

Word of mouth:

- A German LLMD had patients' blood analyzed there in particular for Borrelia
 - All came back negative
 - Many reasons, why test may have „failed“
 - No / not enough Borrelia DNA in blood
 - Unsuitable sample preparation and processing
 - Patients were truly negative

Microbiome Analysis

<https://www.frontiersin.org/journals/microbiology/articles/10.3389/fmicb.2023.1148466/full>

- Analyzed microbiome of neonates using Nanopore sequencing

=> I started a cooperation with a large German lab concerning microbiome analysis via Metagenomics

- Currently using 16S technology
- A suitable Genestrip database has been compiled...

5. The Future

Metagenomics: Many Areas of Application...

- In medicine and diagnostics
 - For hygiene and pathogen surveillance
 - In biology and ecology for species surveillance
 - For quality assurance in food production
 - In farming and agriculture to control plant infections
- Etc.

Metagenomics as a „Disruptive Technology“

My hypothesis:

- Many traditional lab tests will be replaced
 - Labs better adjust early on time...
- Major change and challenge:
Wet lab work will have to be combined with IT-based data analysis
- Similar to replacement of
 - typewriter replaced by PCs (70s),
 - celluloid replaced by digital film (80s and 90s)

Thanks for your interest!
Discussion / Questions... (!?)

Backup Slides

In Reach: Sequencing “At Home”

Via an “all in one” device such as “Traxion”:

<https://nanoporetech.com/products/sequence/traxion>

Mini Study using Genomic Files of People

Background:

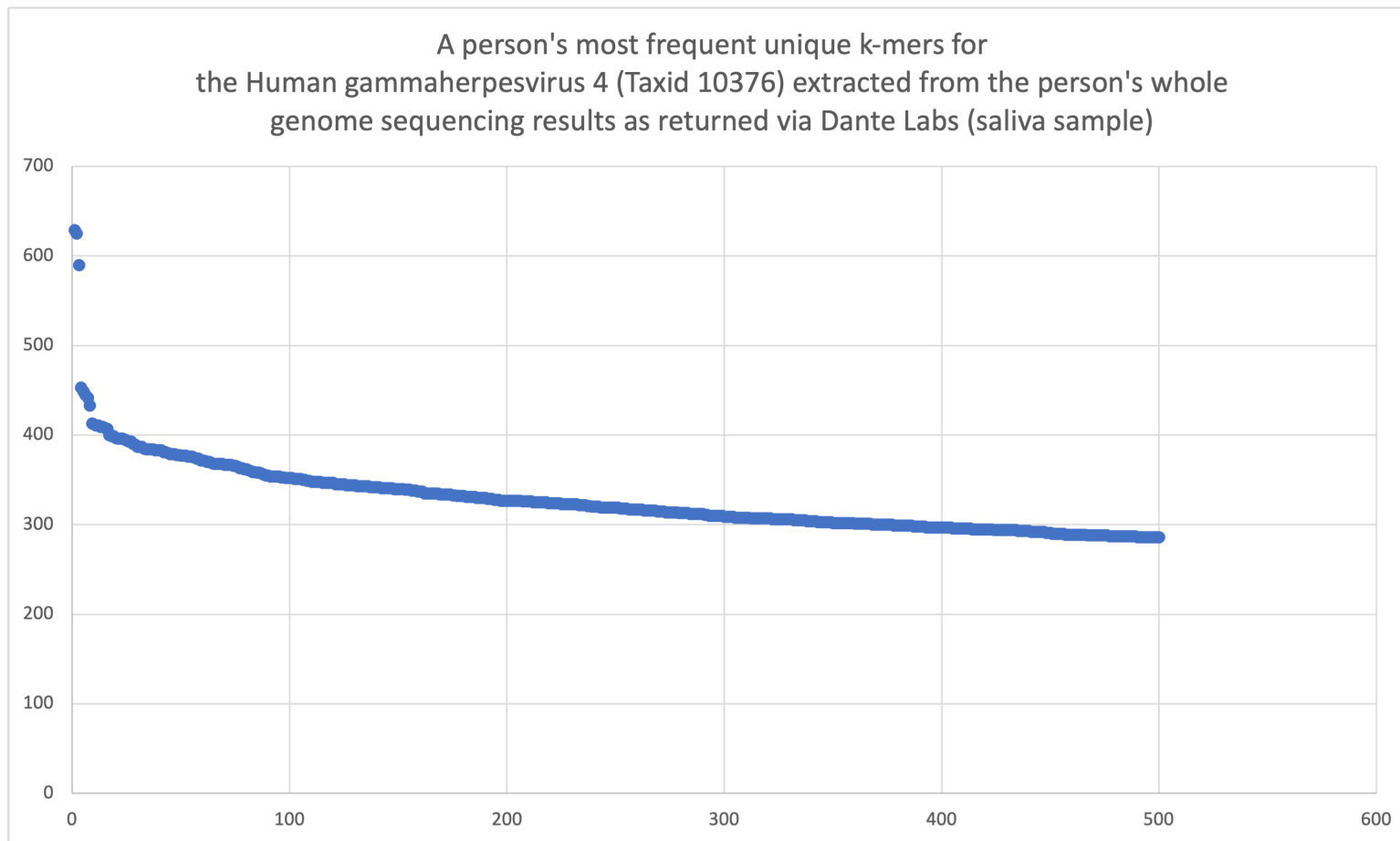
- For > 500 Euros you can order an analysis of your personal genome online
- E.g. you send a saliva sample to a provider and get fastq files (among others) for download

In this context, 16 persons gave their corresponding fastq files for metagenomic analysis

- Personal consent given and ethics assessment passed
- Most users involved were in the „Hannover Lyme Self-Help Group“ ... (Thanks 😊)

Results Summary for Viruses

- All of the „usual“ herpes viruses were detected in accordance with prevalence in general population
 - High and consistent k-mer counts
- One person's sample (only) resulted in 85% genomic coverage of the EBV virus!
 - The person had fatigue symptoms claimed to be consistent with “chronic EBV”
- **So, even though the fastq files were just meant for analyzing the human genome, they allowed for finding viruses typically residing the mouth!**



Another example of viral results for a person (saliva sample, Dante Labs):
<https://genestrip.it.hs-heilbronn.de/gweb/view?lan=en&jobid=92>
(Login Required)

Results Summary for Bacteria

- “Bad Hits” for potentially tick-borne bacteria:
 - k-mer counts were too low for reliable detection
 - ratio of k-mers and unique k-mers was inconsistent
- “Ok Hits” for “Streptococcus Mutans”:
 - High enough k-mer counts
 - But ratio of k-mers and unique k-mers was still inconsistent
- Likely reasons:
 - Sample preparation was unsuitable for preserving bacterial DNA
 - Provider’s sequencing filtered out many of the bacterial reads
 - No such bacteria (Borrelia) in sample (as it was saliva)